

CS 270 – Lab 9 Q1

Question 1: (F a b)

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;input-contract: a and b are nonnegative integers
;output-contract: (F a b) is a recursively defined integer
;
; pattern redacted for students
(define (F a b)
  (if (zero? b) a (+ 1 (F a (- b 1))))))
```

Use Equational Reasoning to prove (F 5 1) evaluates to 6.

Expression	Justification
(F 5 1)	Premise
(if (zero? 1) 5 (+ 1 (F 5 (- 1 1))))	Apply definition of F with a=5 and b=1
(if #f 5 (+ 1 (F 5 (- 1 1))))	Evaluate zero?
(+ 1 (F 5 (- 1 1)))	Evaluate if
(+ 1 (F 5 0))	Evaluate – (subtraction)
(+ 1 (if (zero? 0) 5 (+ 1 (F 5 (- 0 1)))))	Apply definition of F with a=5 and b=0
(+ 1 (if #t 5 (+ 1 (F 5 (- 0 1)))))	Evaluate zero?
(+ 1 5)	Evaluate if
6	Evaluate + (addition)